When Documents Don't Connect: Traceability Burden and Loopback Behavior in GMP Document Navigation

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In pharmaceutical manufacturing environments governed by Good Manufacturing Practice (GMP) regulations, documents form interdependent networks, yet those relationships are typically represented as textual references rather than navigable structures. This paper argues that traceability is not only a property of documentation but an HCI problem about how document structure shapes navigation, and how navigation behavior reveals the structure practitioners are forced to work through. Interface structure determines which document relationships become visible and which remain opaque — making navigation design a tractable target for HCI intervention even within the no-modification constraint. I introduce two design contributions that address this problem from complementary directions: an external traceability visibility prototype that makes document linkage navigable without modifying validated source files, and a behavioral navigation tracking prototype that operationalizes loopback behavior as an empirical indicator of document-structural friction. I report two prototype-based studies with $N = 20$ GMP-experienced participants. Study 1 evaluated a graph-based Traceability Explorer over a synthetic corpus of 150 GMP records, identifying 31 missing CAPA links, 103 broken downstream links, and 21 lifecycle mismatch risks; path reconstruction time reduced from approximately 7 minutes to approximately 4 minutes, with all 20 participants completing target pathways successfully. Study 2 deployed a Cognitive Navigation Explorer during a standardized out-of-specification investigation scenario; loopback behavior was observed across all 20 sessions (mean: approximately 5 loopbacks), with experienced practitioners averaging 2–3 loopbacks and less experienced participants averaging 7–10, while participants consistently attributed navigation difficulty to document structure rather than interface design. The paper makes three contributions: it introduces document traceability as a design problem specific to regulated environments; identifies the no-modification constraint as a novel design boundary for document navigation research; and argues that making document relationships visible redistributes epistemic access by externalizing navigational structure that would otherwise remain implicit, unevenly distributed, and individually absorbed.

Additional Key Words and Phrases: document traceability, loopback behavior, GMP, pharmaceutical manufacturing, navigation friction, epistemic access, regulated environments, invisible work

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1 Introduction

In pharmaceutical manufacturing environments governed by Good Manufacturing Practice (GMP) regulations, documents do not exist in isolation. A single deviation investigation may require navigating across standard operating procedures, analytical method protocols, validation records, change control documents, and specification files — each of which references others through textual cross-references rather than navigable links. When these references are complete and current, traceability is possible. When they are missing, outdated, or structurally ambiguous, the burden of

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reconstructing pathways falls entirely on the practitioner. The practitioner becomes, in effect, the missing infrastructure between documents.

This paper argues that traceability is not only a property of documentation — whether references exist, whether audit trails can be reconstructed — but a condition of interaction. When document relationships are only partially visible, practitioners must navigate through incomplete structures, and their navigation behavior reflects the structure they are working through. A loopback — a return to a previously visited document — is not simply evidence of inefficient searching. It is evidence that the path forward was not yet legible: that the document system did not resolve the judgment demand on the first pass. In information foraging terms [5], a loopback is a rational response to weak information scent, not a navigational error. This reframing matters not only analytically but methodologically. If loopbacks are structural signals rather than usability signals, they can serve as an empirical instrument for document system evaluation without deploying instrumentation inside validated systems.

This shift matters because navigation in regulated environments is consequential. Whether a practitioner can reconstruct a defensible pathway through documents is not only a matter of efficiency. It shapes which records can be brought into view, which procedural relationships can be established, and which justifications can be sustained during investigation and review. When linkage structure is difficult to reconstruct, practitioners do not merely spend more time searching. They work under conditions of partial legibility, relying on memory, inference, and accumulated experience to bridge gaps the system does not make visible. These conditions are not neutral. They distribute navigational burden unequally across practitioners and shape who can move through document systems with confidence. Interface structure determines what practitioners can see and act on — and when that structure fails to externalize document relationships, the burden of reconstruction falls entirely on the practitioner rather than on the system.

I report two prototype-based studies in GMP environments. Study 1 develops and evaluates a Traceability Explorer — an external graph-based visibility layer over a synthetic corpus of 150 GMP records — and demonstrates that making linkage structure navigable reduces path reconstruction burden. Study 2 develops and evaluates a Cognitive Navigation Explorer that records document transitions, visit frequency, and loopback events during a standardized out-of-specification investigation scenario, and demonstrates that loopback frequency varies systematically with experience in ways that reflect document structure rather than individual reasoning quality. Neither prototype modifies validated source documents — a constraint constitutive of GMP-regulated environments.

The contributions of this work are threefold. First, I introduce document traceability as an HCI problem specific to regulated environments, identifying the no-modification constraint as a novel design boundary for document navigation research. Second, I show that behavioral navigation tracking — loopback analysis in particular — can function as an empirical method for surfacing document-structural friction without deployment inside validated systems. Third, I offer a theoretical framing that connects document architecture to navigation behavior to accountability conditions: in GMP environments, whether a practitioner can reconstruct a defensible pathway depends not only on their expertise but on whether the document system makes its own structure legible.

2 Related Work

Documents as Organizational Action

The foundational premise of this study — that document systems do not merely store information but actively organize the conditions under which work can be done — draws on Harper's (1998) ethnographic account of documentation practices at the IMF. Harper shows that documents are not neutral containers: their accessibility, their linkage to

other documents, and the navigational demands they impose on users are constitutive of organizational action. When document relationships are visible and traversable, practitioners can reconstruct accountable pathways; when they are not, that reconstruction work falls to individuals. Applied to GMP environments, this framework implies that traceability is not simply a property of document content but of document architecture — and that architectural failures redistribute labor in ways that are rarely measured.

Articulation Work and the Human as Missing Infrastructure

Schmidt and Bannon [6] introduce the concept of articulation work to describe the coordinative labor that makes other work possible: the activities required to manage the distributed, interdependent nature of cooperative work that formal systems do not support. In regulated document environments, cross-reference reconstruction is a form of articulation work. When linkage structure is incomplete or invisible, practitioners must manually bridge the gaps between documents, inferring relationships from partial cues and memory. This bridging labor is not represented in any quality record, even though the work depends on it.

Information Foraging and Loopback as Structural Signal

Pirolli and Card [5] develop information foraging theory to explain how people navigate information environments: users follow information scent — proximal cues about the value and accessibility of information sources — and move between information patches in ways that maximize gain relative to cost. A loopback, in this framework, is a rational response to weak or misleading scent: when the path forward does not resolve the task, returning to a previously visited document is the appropriate behavior. This reframing has methodological implications. Loopback frequency can be used as a diagnostic instrument for document system evaluation, identifying the specific documents or cross-reference junctions producing the most friction — without deploying instruments inside validated systems. Applied to GMP document navigation, loopbacks indicate that the document system did not provide sufficient scent to support forward navigation, and that the practitioner was forced to reorient from a prior position.

Situated Action and the Conditions of Document Navigation

Suchman's (1987) account of situated action establishes that navigation through a document system is not plan-driven but shaped by the immediate conditions of execution — what is visible, accessible, and interpretable at the moment a decision must be made. When cross-references are broken, outdated, or ambiguous, the conditions of navigation are degraded, and the practitioner must substitute inferential work for structural guidance. This study examines the behavioral traces of that substitution.

Invisible Work and the Unmeasured Labor of Path Reconstruction

Star and Strauss [7] show that work performed in the gaps of formal systems tends to become invisible: it is not captured by indicators, not attributed to the structural conditions that produced it, and not recognized as labor by the organizations that depend on it. In regulated document environments, the manual reconstruction of broken or incomplete traceability pathways is precisely this kind of invisible work. The present study makes this labor visible in two ways: through graph analysis of linkage structure, which identifies where invisible bridging is likely to be required, and through behavioral tracking, which records the navigation patterns that invisible bridging produces.

Prototype-Based Evaluation

The methodological approach builds on established practice for prototype-based evaluation in HCI. Nielsen [4] demonstrates that think-aloud evaluation with small participant samples reliably surfaces usability and navigability problems. Hutchinson et al. [3] establish the logic of using a prototype simultaneously as a design intervention and as an analytic probe — a technology probe makes the conditions of use visible rather than simply testing interface quality.

AI-Assisted Navigation and Interface Effects

Within HCI, work on AI-assisted decision making has established that interface structure actively shapes which information becomes salient — and which remains invisible — to practitioners navigating complex information environments. Wang and Yin [9] demonstrate that explanation interfaces alter how decision makers engage with outputs depending on domain expertise, establishing that navigation interfaces are active shapers of what relationships become visible rather than neutral conduits through which practitioners move freely. Green and Chen [1] show that algorithm-in-the-loop settings systematically constrain the conditions under which independent judgment is possible — a finding that grounds the no-modification constraint's design logic in this study: when interface structure determines what practitioners can see and act on, external visibility layers that preserve interpretive authority are not merely a design preference but a structural requirement of regulated navigation environments.

Prior work has not examined document traceability as a navigation and interaction problem in GMP-regulated environments, nor has it treated loopback behavior as an empirical indicator of document-structural friction. This study addresses both gaps.

3 Method

Study Context

This study was conducted in the context of pharmaceutical manufacturing quality systems operating under GMP regulations. GMP document environments are characterized by interdependent document networks — SOPs, CAPA records, analytical method protocols, validation artifacts, and specification documents — in which cross-references are textual rather than navigable, and the burden of reconstructing pathways falls on practitioners.

Participants

Both studies were conducted with the same participant pool of $N = 20$, recruited from the same pharmaceutical quality team. All participants had GMP work experience with daily procedural document use; most had direct experience with deviation or out-of-specification investigation workflows. The participant pool included practitioners across a range of experience levels: 3 participants had limited GMP investigation experience (less than 2 years), 14 had intermediate experience, and 3 had extensive experience (5 or more years). This sample size is consistent with established think-aloud methodology, in which $N = 20$ expert participants in a domain-specific population reliably surfaces dominant usability and friction patterns [4]. The same-sample design was a deliberate methodological choice to eliminate between-participant variance in domain knowledge as a confound, prioritizing internal consistency over sample independence. Recruiting the same participants for both studies ensured consistent baseline familiarity with GMP document conventions. This design choice introduces the possibility of shared response tendencies across evaluations — a limitation acknowledged below.

Synthetic Document Corpora

Both studies used synthetic document corpora constructed in Python to reflect the structural and terminological characteristics of real GMP documentation without reproducing proprietary content. The no-modification constraint that defines the core design boundary of this work applies equally to synthetic and live corpora in GMP environments.

Study 1: Traceability Explorer

Document Network Analysis. The Study 1 corpus represented an interdependent network of 150 GMP records — SOPs, CAPA records, deviation reports, change control documents, and specification documents — with explicit cross-references and lifecycle state assignments. The corpus was modeled as a directed graph using Python and NetworkX, with documents as nodes and cross-references as directed edges. The graph analysis identified three categories of linkage problem: missing links (31 deviations with no explicit CAPA connection), broken downstream links (103 records with no change-control linkage), and lifecycle mismatches (21 outdated references, lifecycle conflicts, and duplicates).

Prototype Development. Based on the network analysis, I developed an external Traceability Explorer as a Streamlit web application organized into three tabs (Figures 1–3):

- **Chain Viewer** (Figure 1): presents the four-record documentation chain (Deviation → CAPA → Change Control → SOP Reference) for any selected record, with missing and broken links flagged inline.
- **Risk Summary** (Figure 2): surfaces traceability risks at the dataset level — unlinked deviations, broken downstream links, outdated SOP references, and lifecycle mismatches — with detail tables for each risk category.
- **Record Explorer** (Figure 3): allows filtering across the full 150-record corpus by record type, status, and product, surfacing linkage gaps as observable patterns.

The system operates over the metadata structure; it does not modify, version, or interact with the synthetic source documents.

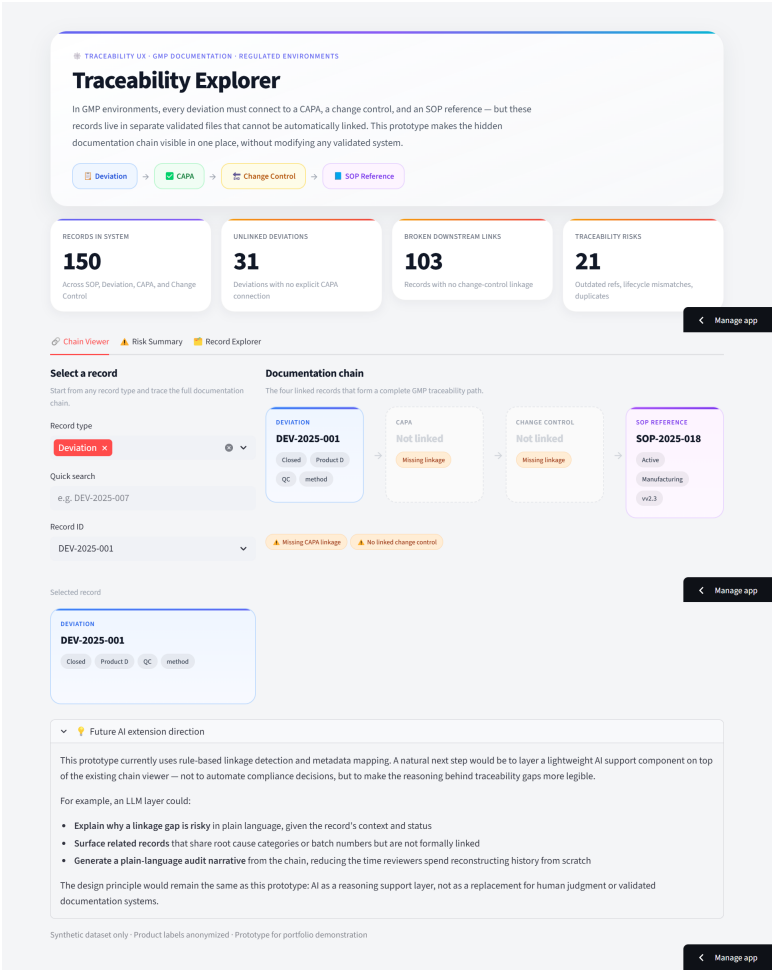


Fig. 1. Traceability Explorer: Chain Viewer tab, showing the documentation chain for DEV-2025-001 (Deviation → CAPA → Change Control → SOP Reference). CAPA and Change Control nodes are marked "Not linked" with missing linkage warnings, while SOP-2025-018 is resolved as active and current. The four-record chain structure makes the full traceability pathway visible in one place without modifying any validated system.

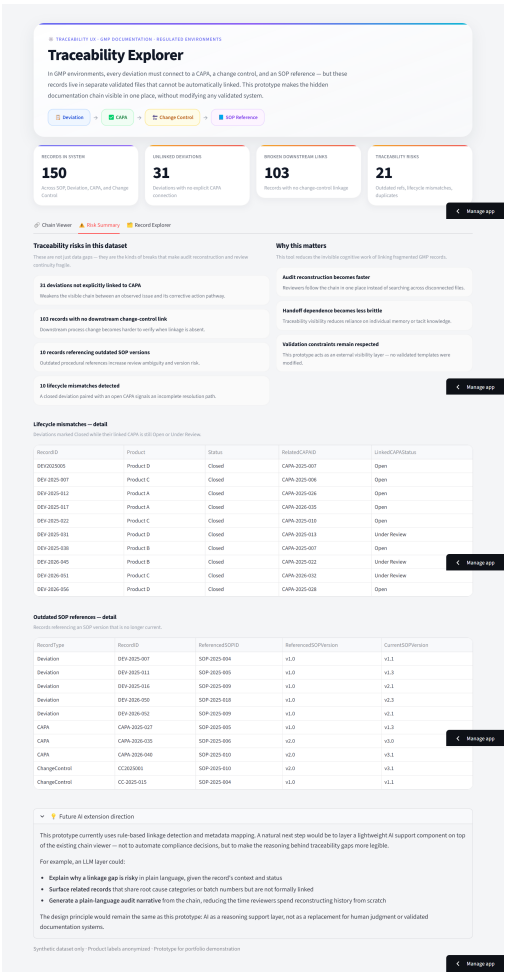


Fig. 2. Traceability Explorer: Risk Summary tab, showing four risk categories with explanations of why each matters for audit reconstruction and review continuity. Lifecycle mismatches detail table shows 10 closed deviations whose linked CAPAs remain open or under review. Outdated SOP references detail table shows 10 records referencing superseded SOP versions.

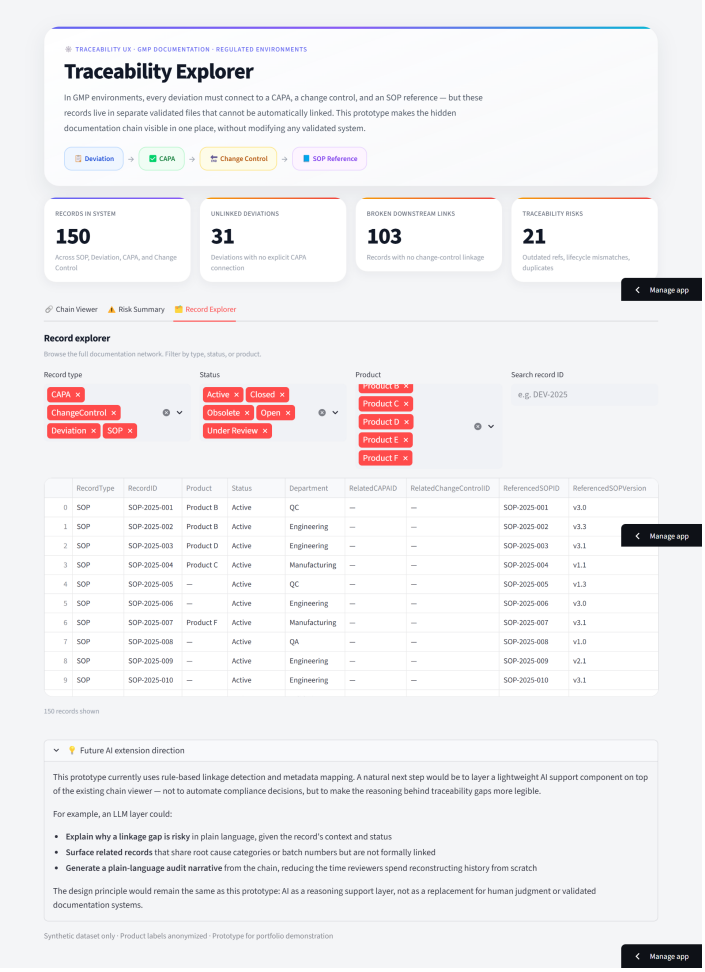


Fig. 3. Traceability Explorer: Record Explorer tab, showing the full 150-record corpus with filters for record type, status, and product. The table surfaces linkage gaps as observable patterns across the dataset — missing CAPA IDs and change control IDs are visible as empty cells.

Evaluation. Participants completed three task-based traceability scenarios: locating the next downstream document in a reference chain, determining whether a given pathway was complete, and identifying where a reference had become outdated or broken. Post-task measures included task completion time, path completion accuracy, document discovery rate, ease of use, and adoption intent.

Study 2: Cognitive Navigation Explorer

Study 2 follows directly from Study 1's finding that document linkage failure is structurally widespread and creates measurable navigation burden. If linkage gaps produce reconstruction costs when they are made visible, they should also produce observable behavioral traces when practitioners navigate without structural support. Study 2 tests this implication directly: rather than measuring the effect of visibility improvements, it tracks the navigation behavior that

emerges when practitioners work through an investigation scenario without traceability support, treating loopback patterns as empirical evidence of structural friction.

Scenario Design and Prototype. Participants were given a standardized out-of-specification investigation scenario — an ELISA assay result exceeding specification — and asked to navigate across seven synthetic GMP documents to identify the relevant procedures and supporting records (Figure 5). No navigation hints were provided. The Cognitive Navigation Explorer (Figure 4) recorded document-to-document transitions with action type (forward navigation, cross-reference jump, or loopback), visit count and time spent per document, and the full navigation sequence as a directed path. Results were displayed in a Cognitive Navigation Analysis dashboard (Figure 6) showing the navigation path network, document visit counts, average time per document, and the full navigation sequence log.

A loopback was operationally defined as any return to a document that had already been visited within the same session, where the intervening path had visited at least one other document — excluding immediate back-navigation while capturing returns that indicate unresolved judgment demand.

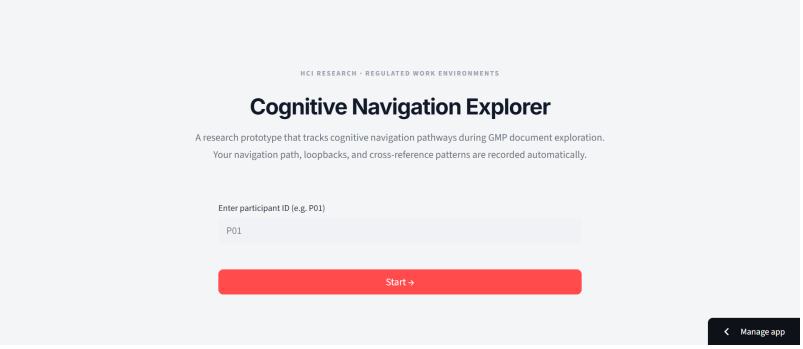


Fig. 4. Cognitive Navigation Explorer: landing screen, showing the study description and participant ID entry. The interface makes explicit that navigation paths, loopbacks, and cross-reference patterns are recorded automatically.

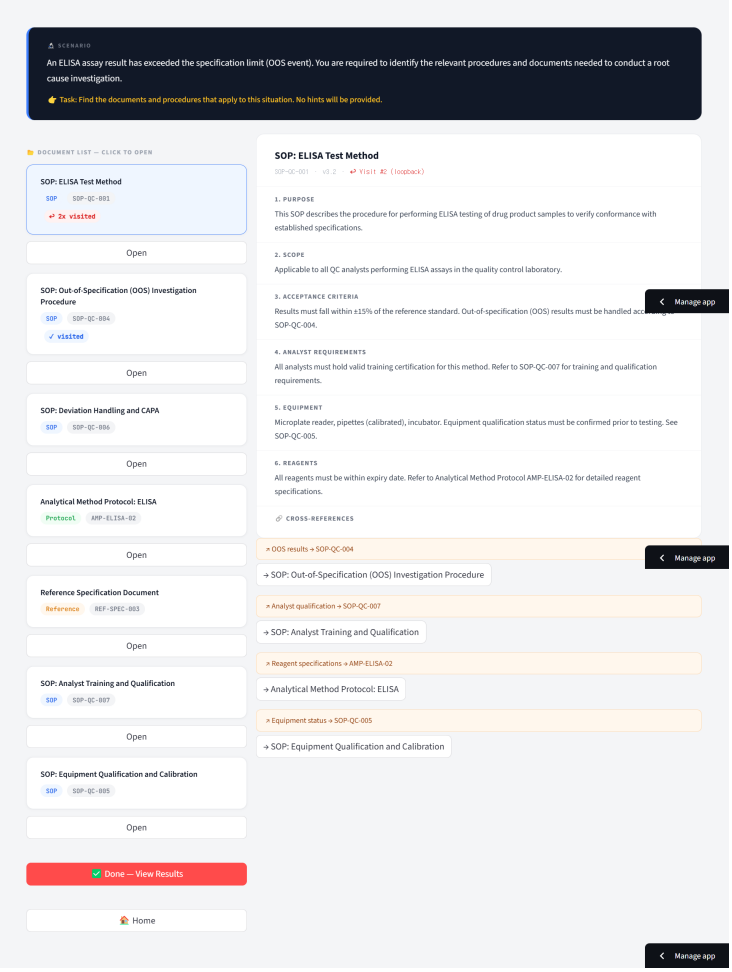


Fig. 5. Cognitive Navigation Explorer: OOS investigation scenario in progress. The ELISA Test Method SOP is open at Visit #2 (loopback), indicated by the "2x visited" badge on the document list and the "Visit #2 (loopback)" label in the document header. The cross-references section shows five linked documents with highlighted links to OOS Procedure, Analyst Training, Reagent Protocol, and Equipment Qualification.

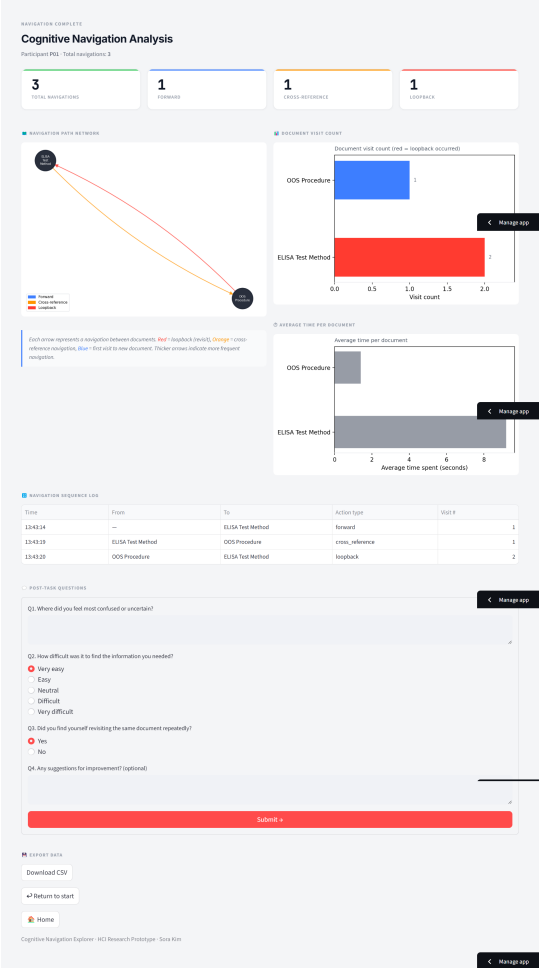


Fig. 6. Cognitive Navigation Analysis: results dashboard for Participant P01, showing 3 total navigations (1 forward, 1 cross-reference, 1 loopback). The navigation path network (left) shows the ELISA Test Method as the hub node with arrows color-coded by action type (blue = forward, orange = cross-reference, red = loopback). The document visit count chart (right) shows ELISA Test Method visited twice (red bar = loopback occurred) and OOS Procedure visited once.

Evaluation. Post-task measures included document list clarity, content readability, cross-reference clarity, navigation awareness, and real-world utility. Qualitative feedback probed the sources of navigation difficulty — specifically, whether participants attributed confusion to the prototype interface or to the structure of the documents themselves.

4 Findings

Study 1: Traceability Visibility

Linkage Visibility as an Interaction Condition. The most consequential participant response to the Traceability Explorer concerned personnel handover — a condition that occurs frequently in pharmaceutical quality operations and that current document systems handle poorly. When a quality practitioner changes roles or leaves a team, the change control

records, deviation reports, and CAPA documents they managed do not transfer with any navigable account of their relationships. The incoming practitioner must reconstruct the full history of interconnected quality events manually. Participants described this reconstruction work as one of the most time-consuming and error-prone aspects of their roles. As one participant put it: "every time someone leaves the team, the first three months of their replacement is basically archaeology."

What matters here is not only that reconstruction takes time, but that the system fails to externalize relationship structure in a form practitioners can act on. The Traceability Explorer addressed this directly by making cross-reference relationships visible as a navigable chain, allowing a documentation history that would otherwise remain fragmented across records to be seen at once. In this sense, traceability visibility changed more than task speed. It changed the interaction conditions under which practitioners could reconstruct, compare, and justify pathways through the record set.

Task Performance. In task-based evaluation, the Traceability Explorer was associated with a reduction in path reconstruction time from approximately 7 minutes to approximately 4 minutes — a 30–60% reduction consistent across task types and experience levels. All 20 participants successfully completed target pathway reconstruction, and all 20 indicated adoption intent. Ease of use averaged 5.0 out of 5, with document discovery rates and broken-reference detection consistently high across sessions.

These findings matter not only as indicators of successful interface use, but as evidence that rendering linkage structure visible reduces the amount of inferential bridging practitioners must do in order to reconstruct an accountable pathway. The prototype did not alter any underlying documents. It altered what relationships became immediately legible.

Study 2: Behavioral Navigation Tracking

Loopback as a Structural Signal. Loopback behavior was observed across all 20 participant sessions, with a mean of approximately 5 loopbacks per session. The most common pattern was: Analyst Training Record → ELISA Test Method SOP → Method Validation Report → ELISA Test Method SOP → OOS Investigation Procedure. The return to the ELISA SOP is not evidence of confusion in a generic sense. It is evidence that the first visit did not fully resolve the judgment demand — that the document's cross-references to adjacent procedures required a second visit to evaluate completely.

Interface usability ratings were consistently positive across all participants (5.0/5 across all post-task measures). Less experienced participants specifically identified SOP structure — ambiguous cross-references, non-obvious sequencing, overlapping scope across documents — as the source of navigational uncertainty. This dissociation between interface usability and navigation difficulty supports the central methodological claim of the paper: loopback behavior is a signal of document-structural friction, not interface friction.

Experience and Navigation Strategy. Participants with five or more years of GMP investigation experience averaged 2–3 loopbacks per session and typically opened a document related to analyst performance first — reflecting the practical heuristic that pipetting errors and reagent preparation are common sources of ELISA out-of-specification results. Intermediate-experience participants averaged 4–6 loopbacks. Less experienced participants averaged 7–10 loopbacks and were more likely to open the reference specification first, attempting to reconstruct the investigative pathway from structural cues rather than from internalized practice knowledge.

This experience asymmetry has direct implications for document system design. As one less experienced participant noted: "I kept going back to the ELISA SOP because I wasn't sure if I'd missed something — there were so many cross-references and I couldn't tell which ones actually mattered for this situation." The navigation burden that experienced practitioners have reduced through years of practice remains fully present for less experienced practitioners — and is borne entirely by the practitioner, not redistributed or reduced by the document system.

5 Discussion

Cross-Study Synthesis

Together, the two studies establish a structural-behavioral relationship. Study 1 shows that document linkage is incomplete and that making it visible reduces reconstruction burden. Study 2 shows that practitioners respond to incomplete linkage through navigation patterns — particularly loopbacks — that are consistent across sessions and reflect document structure rather than interface design or individual error. Traceability failure does not remain at the level of documentation. It propagates into behavior, and that behavioral trace is empirically observable.

Both studies also converge on a finding about epistemic inequality within the document environment. The practitioners who bear the greatest navigation burden are those with the least accumulated implicit knowledge of the system. Less experienced participants averaged more than three times as many loopbacks as experienced practitioners — not because they reasoned poorly, but because they lacked the internalized map of the document system that experienced practitioners have developed over years. Making that structure visible is therefore not only a usability improvement. It is a redistribution of epistemic access.

Traceability as an Interaction Problem

The central contribution of this study is a reframing: traceability in regulated environments is not only a documentation property but a condition of interaction. When document relationships are only partially visible, the burden of reconstructing pathways is redistributed to practitioners, and that redistribution is not neutral. It falls disproportionately on those with the least accumulated implicit knowledge of the document system. Making linkage structure visible does not merely improve efficiency; it changes who bears the cost of structural incompleteness and who can navigate an accountable pathway with confidence.

This extends Harper's (1998) account of documents as organizational action. The present study shows what happens when those linkage conditions fail: the articulation work of bridging incomplete document networks becomes individually absorbed, invisible to quality systems, and unequally distributed across a workforce stratified by experience.

Loopback as Method

The methodological contribution of this study is the operationalization of loopback behavior as an empirical indicator of document-structural friction. Prior work on navigation behavior has often treated loopbacks primarily as usability signals — evidence that an interface is confusing. The present study offers a different interpretation, grounded in information foraging theory [5]: a loopback is a rational response to weak information scent, not a navigational error. When a document does not provide sufficient cues to support forward navigation, returning to a prior document is the appropriate behavior.

This reframing has methodological implications beyond the present study. Loopback frequency and the location of reorientation events are tractable metrics that can be collected through external prototypes and used to identify

structural friction before it is absorbed as individual burden — without deploying instruments inside validated systems. The mean of approximately 5 loopbacks per session observed here, and the systematic variation by experience level, suggest that loopback frequency is a sensitive metric for this purpose.

Articulation Work and the Invisibility of Navigation Burden

The handover finding illustrates the mechanism that Schmidt and Bannon [6] describe as articulation work. When a quality practitioner changes roles, the document relationships they have accumulated do not transfer. The incoming practitioner must reconstruct these relationships manually — real labor that is invisible to any quality metric and unattributed to the structural conditions that produce it. The Traceability Explorer makes this invisible labor visible by externalizing the relationship structure that practitioners currently hold in implicit knowledge.

Epistemic Access in Regulated Work

The convergence of findings across both studies points toward a broader framing: document system design in regulated environments is, among other things, a problem of epistemic access — of who can reconstruct defensible pathways, under what conditions, and at what cost. The experience asymmetry documented in Study 2 is not a training gap. It is a design gap. Experienced practitioners have internalized a map of the document system that the system itself does not provide. Making linkage structure visible is a way of externalizing that map — of building into the document environment the navigational support that currently exists only in the heads of experienced practitioners.

Design Implications for Regulated Document Environments

Several design implications follow. First, traceability interventions in regulated environments should be understood not simply as audit-support utilities but as visibility layers that reshape the conditions of navigation. Second, linkages that remain textual but not navigable systematically shift burden onto practitioners, especially in handover conditions and unfamiliar investigative contexts. Third, behavioral traces such as loopbacks can function as design diagnostics: they reveal where document systems fail to provide sufficient structural guidance even when an interface is otherwise usable. Finally, design for regulated document environments must work within a constraint that most navigation research does not face: the underlying source documents often cannot be modified, meaning that navigational support must be layered externally rather than embedded directly into the validated system.

Limitations and Future Work

The most important limitation of this study is the use of synthetic corpora. Both studies used document sets generated to reflect real GMP structures but lacking the full terminological variance, document volume, and cross-reference complexity of live archives. The experience distribution in the participant pool limits the statistical power of experience-stratified comparisons. Future work should deploy both prototypes against live document systems in production quality environments. The handover use case identified in Study 1 is a particularly promising deployment scenario: personnel transitions are frequent, the navigation burden they impose is well recognized by practitioners, and a traceability visibility layer addresses a problem that has no current structural solution. Future work should also examine whether loopback distributions can predict specific classes of structural failure — for example broken references, overlapping document scope, or ambiguous sequencing — across different regulated domains.

6 Conclusion

This paper introduced document traceability as a condition of interaction in regulated environments — not only a documentation property, but a structural determinant of who can navigate defensible pathways, at what cost, and under what conditions. When cross-reference relationships are invisible, the labor of reconstructing them falls to individual practitioners, distributed unequally across a workforce stratified by experience. Making linkage structure visible does not merely reduce search time; it redistributes epistemic access.

Two prototype-based studies establish this claim empirically. Study 1 demonstrated that a graph-based Traceability Explorer reduced path reconstruction time from approximately 7 minutes to approximately 4 minutes; all 20 participants completed target pathways successfully and expressed adoption intent, with particular enthusiasm for the personnel handover use case. Study 2 demonstrated that loopback behavior was observed across all 20 sessions, with a mean of approximately 5 loopbacks per session: experienced practitioners averaged 2–3 loopbacks, intermediate participants 4–6, and less experienced participants 7–10, while participants attributed navigation difficulty to document structure rather than interface design. The experience asymmetry in loopback frequency is not a training finding. It is a design finding: the navigational support that experienced practitioners have internalized through years of practice is not currently built into the document system itself.

Three contributions follow. First, document traceability is introduced as a first-class HCI design problem in regulated environments, where the no-modification constraint creates a design boundary with no precedent in existing navigation research. Second, loopback tracking is proposed as an empirical method for surfacing document-structural friction without deployment inside validated systems. Third, the theoretical framing — traceability as a condition of epistemic access — connects HCI concerns about document navigation to broader organizational questions about how accountability is structured in regulated sociotechnical work.

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